JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY, NOIDA

BTech Semester-6

Minor Project 2

Title:- MULTI-SCALE ENERGY STRESS ANALYSIS

(Integrating Grid Operations and Building Behavior in Southern California)

Submitted By :-

S. No.	Name	Enrollment No.
1.	Hitanshi Bans	23103279
2.	Harshil Chawla	23103280
3.	Shubhi Gupta	23103302

Under the supervision of :-

Dr. Somya Jain

INDEX

1. Introduction 2

2. Motivation 3

3. Problem Statement 4

4. Proposed Framework 5

5. Workflow 7

6. Data Preprocessing Pipeline 8

7. Results and Preliminary Analysis 9

   7.1 Grid Stress Detection using Statistical Thresholding (Z-Score)

   7.2 Grid Stress Detection using Machine Learning (Isolation Forest)

   7.3 Frontend Interface and Visualization Snapshots

8. Conclusion 20

9. References 21

1. INTRODUCTION

Electric power systems are undergoing a major transformation as countries transition toward cleaner and more sustainable energy sources. Renewable technologies such as solar and wind are expanding rapidly and play a key role in reducing carbon emissions. However, unlike traditional power plants, renewable generation is inherently variable and depends heavily on environmental conditions such as sunlight, wind patterns, and temperature. This variability introduces new operational challenges for electricity grids that must continuously maintain a balance between power generation and demand.

At the same time, global electricity demand is increasing due to emerging factors such as the rapid expansion of digital infrastructure and artificial intelligence computing facilities, which require large amounts of energy. In addition, rising global temperatures and frequent heatwaves are increasing cooling demand, particularly in developing countries where air-conditioning adoption is growing rapidly. These trends place additional pressure on power systems and make grid management more complex.

One important challenge in modern grids is managing rapid changes in net load, which represents the electricity demand that must be met by controllable generation after accounting for renewable production. Sudden variations in renewable output or spikes in demand can create periods of grid stress, requiring quick operational adjustments to maintain system stability.

Most existing studies analyze these stress conditions primarily from a grid operations perspective, focusing on renewable forecasting or system-level demand patterns. However, electricity demand is heavily influenced by building energy consumption, particularly through heating, ventilation, and air-conditioning (HVAC) usage. Despite this strong relationship, relatively few studies examine how building-level energy behavior interacts with grid-level stress conditions.

This project, titled "Multi-Scale Energy Stress Analysis: Integrating Building Behavior and Grid Operations in Southern California," proposes a data-driven framework to analyze such stress events. The project will study operational grid data to identify abnormal conditions and examine how renewable variability, weather factors, and demand patterns contribute to grid stress. In later stages, the analysis will incorporate building energy consumption data to investigate how building demand patterns align with these critical periods.

By integrating grid-level operational analysis with building-level energy behavior, the project aims to provide insights that can support better grid planning, demand response strategies, and more resilient energy system management.

2. MOTIVATION

Recent global trends highlight growing pressure on modern electricity grids due to both rising demand and changing environmental conditions. The rapid expansion of digital infrastructure, particularly large-scale data centers and artificial intelligence computing facilities, has significantly increased electricity consumption in several regions. These facilities require continuous and high-density power supply, placing additional stress on already complex power systems.

At the same time, climate change is intensifying heatwaves and increasing cooling demand across many parts of the world. Countries such as India are expected to experience higher electricity demand as air-conditioning adoption grows in response to rising temperatures. Even a small increase in cooling penetration can lead to significant spikes in electricity demand during extreme heat periods.

These trends highlight the need for better analytical tools to understand how electricity demand, renewable variability, and environmental factors interact to create periods of grid stress. Developing data-driven frameworks to identify and analyze such conditions can help grid operators anticipate potential stress events and design more effective demand management and energy planning strategies.

3. PROBLEM STATEMENT

Modern electricity grids must maintain a continuous balance between power generation and electricity demand. However, increasing renewable energy integration and changing demand patterns make it difficult to identify periods when the grid may experience operational stress. Variability in renewable generation, weather conditions, and sudden changes in demand can create complex system behavior that is not always easy to detect or analyze using simple monitoring approaches.

Most existing analyses focus primarily on grid-level operational data or renewable generation patterns. However, electricity demand is strongly influenced by building energy consumption, particularly through cooling and HVAC usage. The interaction between grid-level conditions and demand-side behavior is not always examined together.

Therefore, the main problem addressed in this project is to analyze grid stress conditions using operational data and examine how demand-side factors, particularly building energy consumption patterns, may relate to these periods of stress. The goal is to better understand the factors associated with grid stress and provide insights that could support improved demand management and grid planning.

4. PROPOSED FRAMEWORK

The project follows a multi-stage analytical framework to study grid stress and its relationship with building energy consumption.

Phase 1: Dataset Inspection, Preprocessing, and Feature Engineering

- Two primary datasets are used in the project: a grid operations dataset (2019–2022, originally sampled at 5-minute intervals) and a building energy dataset (2018–2023, hourly resolution), both representing energy systems in California.
- Initial dataset inspection was performed to validate the data structure, check for missing values, duplicates, and abnormal entries. Column names were standardized and basic cleaning operations were performed.
- The grid dataset was resampled from 5-minute resolution to 15-minute intervals to make temporal analysis more manageable while still preserving high-frequency grid dynamics.
- Feature engineering was then performed to create additional explanatory variables. Important engineered features include net load, ramp rates, lag variables, rolling statistics, and cyclical time encoding for hour and month to capture temporal patterns.
- Finally, feature selection was conducted using correlation analysis and domain knowledge to remove redundant variables, reducing the feature set from approximately 70 engineered features to 48 relevant features used for further analysis.

Phase 2: Grid Stress Detection and Analysis

- In this phase, anomaly detection techniques are used to identify periods when the electricity grid experiences abnormal or high-stress operational conditions.
- A baseline statistical approach using Z-score thresholding is applied to detect extreme values in key operational variables such as net load and net load ramp.
- To complement this statistical baseline, a machine learning–based anomaly detection model (Isolation Forest) is implemented to detect complex abnormal patterns in the grid dataset. Prior to training the model, feature selection and dimensionality reduction techniques are used to identify the most relevant variables.
- The anomalies detected through these approaches are then analyzed using exploratory data analysis and may be further grouped using clustering techniques to understand different types of grid stress patterns.
- In addition to that, the framework also explores forecasting-based anomaly detection of net load using machine learning models such as Random Forest, XGBoost and LSTM.
- By comparing predicted net load values with actual observations, large deviations can be used as indicators of abnormal system behavior. This forecasting-based approach provides a potential early indication of emerging grid stress conditions.

Phase 3: Grid Regime Detection and Renewable Variability Analysis

- In this phase, unsupervised learning techniques are used to identify different operating regimes of the electricity grid. These regimes represent recurring system conditions such as normal operation, high renewable generation periods, or high-demand stress periods.
- Clustering methods are applied to key operational variables such as demand levels, renewable penetration, net load behavior, and price indicators to group similar system states.
- The resulting clusters are analyzed to characterize typical grid conditions and understand how system behavior changes across different operational regimes.
- This phase provides a higher-level understanding of how weather conditions, renewable variability, and demand behavior interact to create different grid operating regimes.

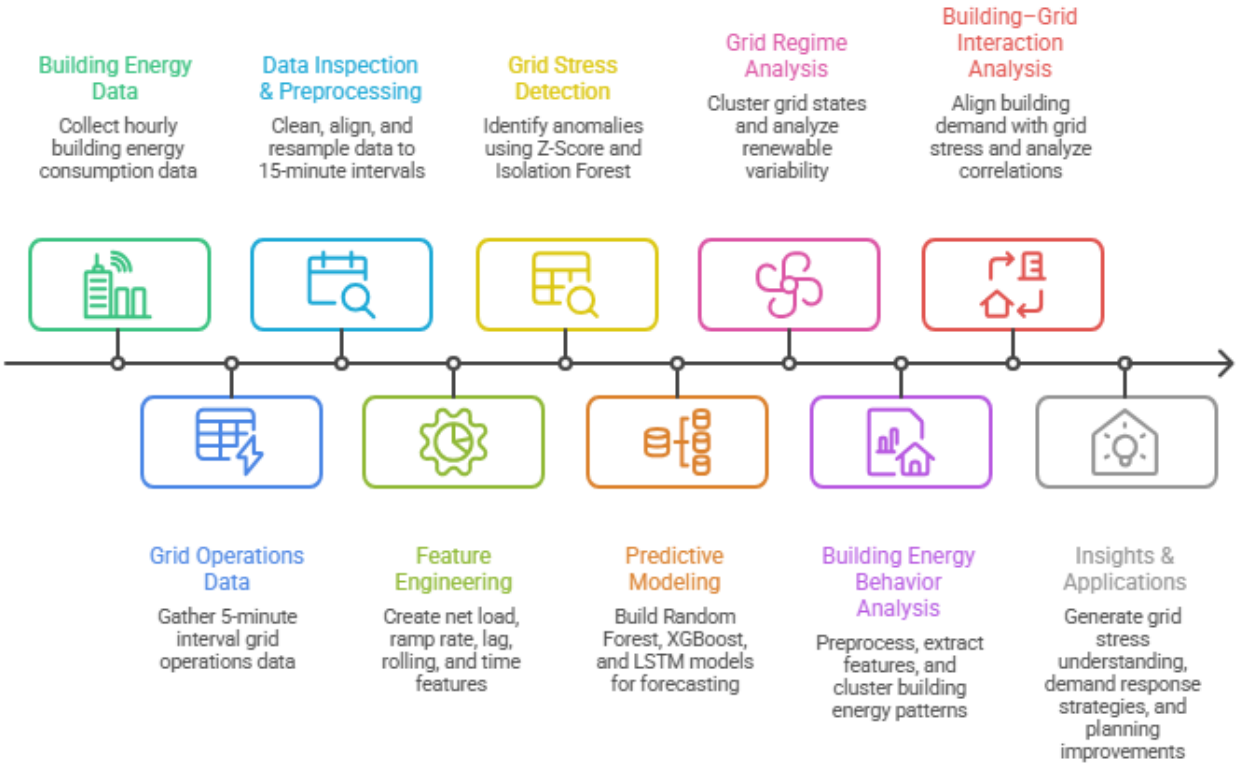
Phase 4: Building Energy Behavior Analysis

- In this phase, building energy consumption data is analyzed to understand how different types of buildings contribute to electricity demand patterns.
- The building dataset first undergoes data cleaning, validation, and basic preprocessing, including handling missing values and preparing relevant consumption variables for analysis.
- Additional features may be engineered to better describe building energy behavior, such as energy intensity indicators or temporal usage characteristics.
- Unsupervised clustering techniques are then applied to group buildings into clusters based on their energy consumption patterns and operational characteristics.
- Cluster validation techniques and exploratory analysis are used to ensure that the resulting clusters represent meaningful building archetypes.
- Each cluster is then analyzed to understand typical energy usage behavior, peak demand patterns, and differences between building categories.
- This phase helps identify how different types of buildings contribute to electricity demand and provides insight into demand-side patterns that may influence grid stress conditions.

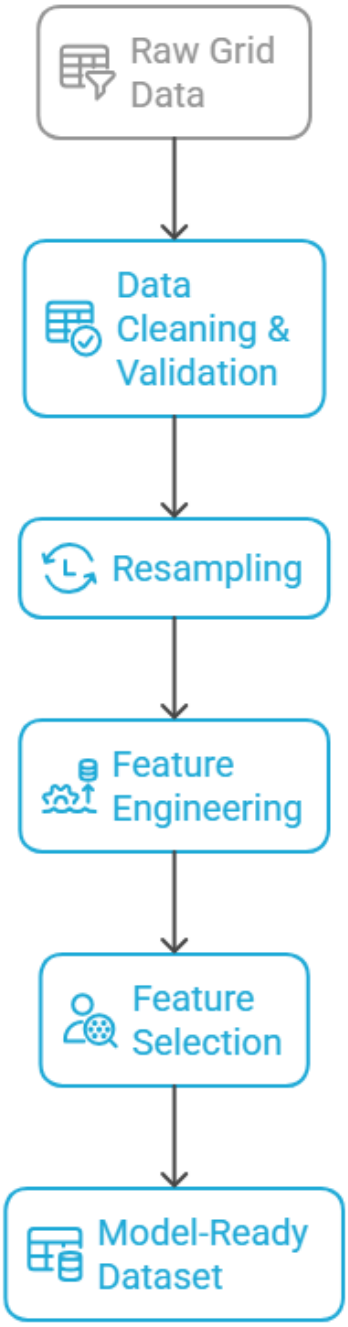
Phase 5: Building – Grid Interaction Analysis

- In the final phase, the relationship between building energy consumption patterns and grid stress conditions is analyzed by aligning building demand data with the detected grid stress periods.
- Electricity demand from different building clusters is aggregated and compared with time periods where grid anomalies or stress events are detected.
- Correlation and comparative analysis are performed to understand how different building archetypes contribute to overall electricity demand during stressed grid conditions.
- This phase aims to provide insights into how demand-side behavior interacts with grid-level dynamics, helping identify opportunities for demand response strategies and improved energy system management.

5.WORKFLOW


Building Energy Data
Collect hourly building energy consumption data
Data Inspection & Preprocessing
Clean, align, and resample data to 15-minute intervals
Grid Stress Detection
Identify anomalies using Z-Score and Isolation Forest
Grid Regime Analysis
Cluster grid states and analyze renewable variability
Building–Grid Interaction Analysis
Align building demand with grid stress and analyze correlations
Grid Operations Data
Gather 5-minute interval grid operations data
Feature Engineering
Create net load, ramp rate, lag, rolling, and time features
Predictive Modeling
Build Random Forest, XGBoost, and LSTM models for forecasting
Building Energy Behavior Analysis
Preprocess, extract features, and cluster building energy patterns
Insights & Applications
Generate grid stress understanding, demand response strategies, and planning improvements

6. DATA PREPROCESSING PIPELINE



7. RESULTS AND PRELIMINARY ANALYSIS

7.1 Grid Stress Detection using Statistical Thresholding (Z-Score)

Table 7.1.1: Comparison of Grid Conditions During Normal Periods and Detected Stress Events

Metric	Normal Average	Shock Event Average	Percent Change
Net Load (mw)	19,250	37,148	+93%
Solar Generation (mw)	3,558	1,110	-69%
Temperature (℃)	17.47	25.64	+46.7%
Net Load Ramp (mw)	0.005	1,757	Huge Spike

Key Insights:

• **Net Load Surge:** During shock events, the average net load rises from 19,250 MW to 37,148 MW (~93% increase), indicating extremely high demand pressure on the grid.

• **Solar Generation Drop:** Solar generation falls by about 69% during stress events, suggesting that reduced renewable output contributes significantly to grid stress.

• **Temperature:** Average temperature during shock events is about 46% higher, indicating that heat-driven electricity demand (e.g., cooling load) may play an important role.

• **Ramp Surge:** Net load ramp increases sharply during shock periods, indicating sudden spikes in demand or rapid loss of renewable generation, both of which create challenges for grid stability.

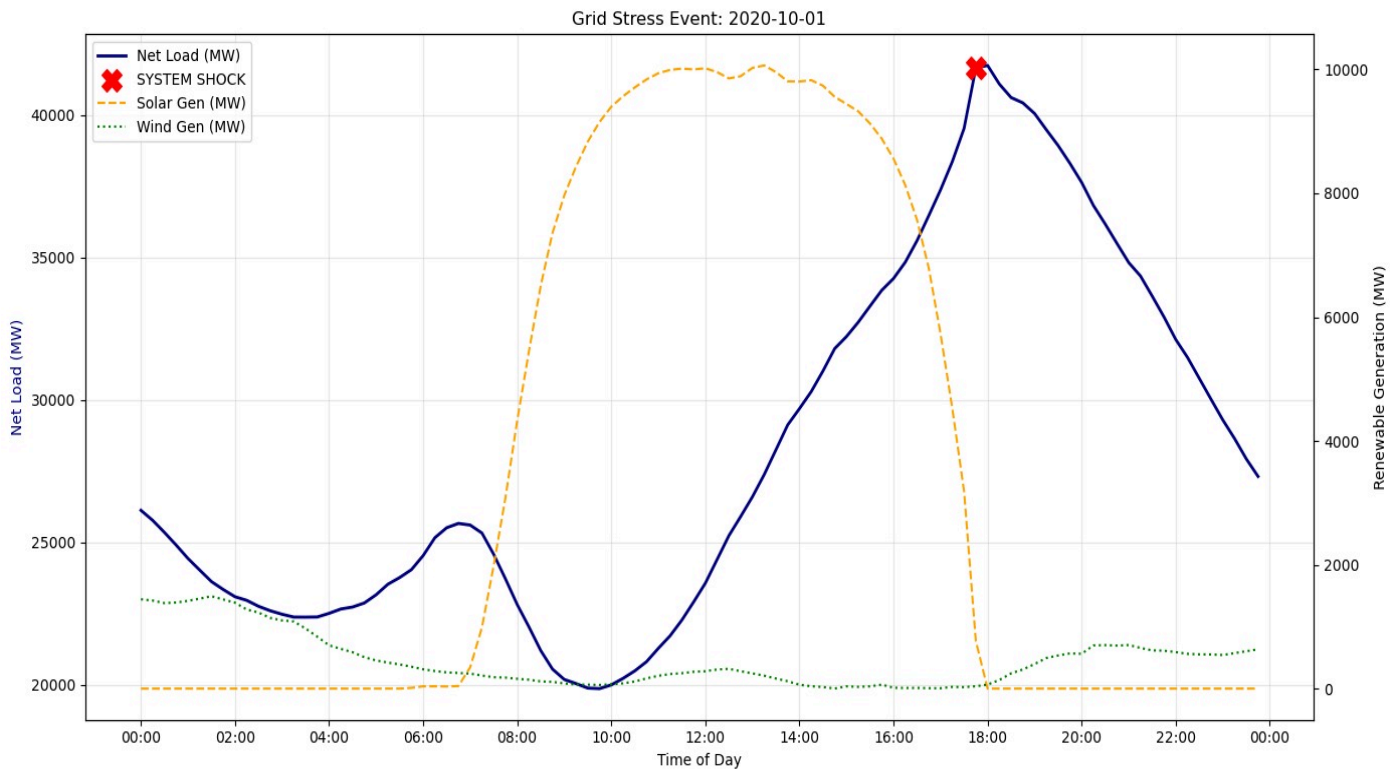


Figure 7.1.1: Net load and renewable generation profile during a detected grid stress event (1 October 2020).

<u>Key Insights:</u>

• **Evening Net Load Peak:** Net load rises steadily through the afternoon and reaches its maximum around 18:00.

• **Solar Generation Drop:** Solar generation declines sharply in the late afternoon, reducing available renewable supply and increasing the need for conventional sources of generation.

• **Renewable-Demand Mismatch:** As solar output falls while electricity demand continues to rise, the net load increases rapidly, ultimately resulting in high grid stress.

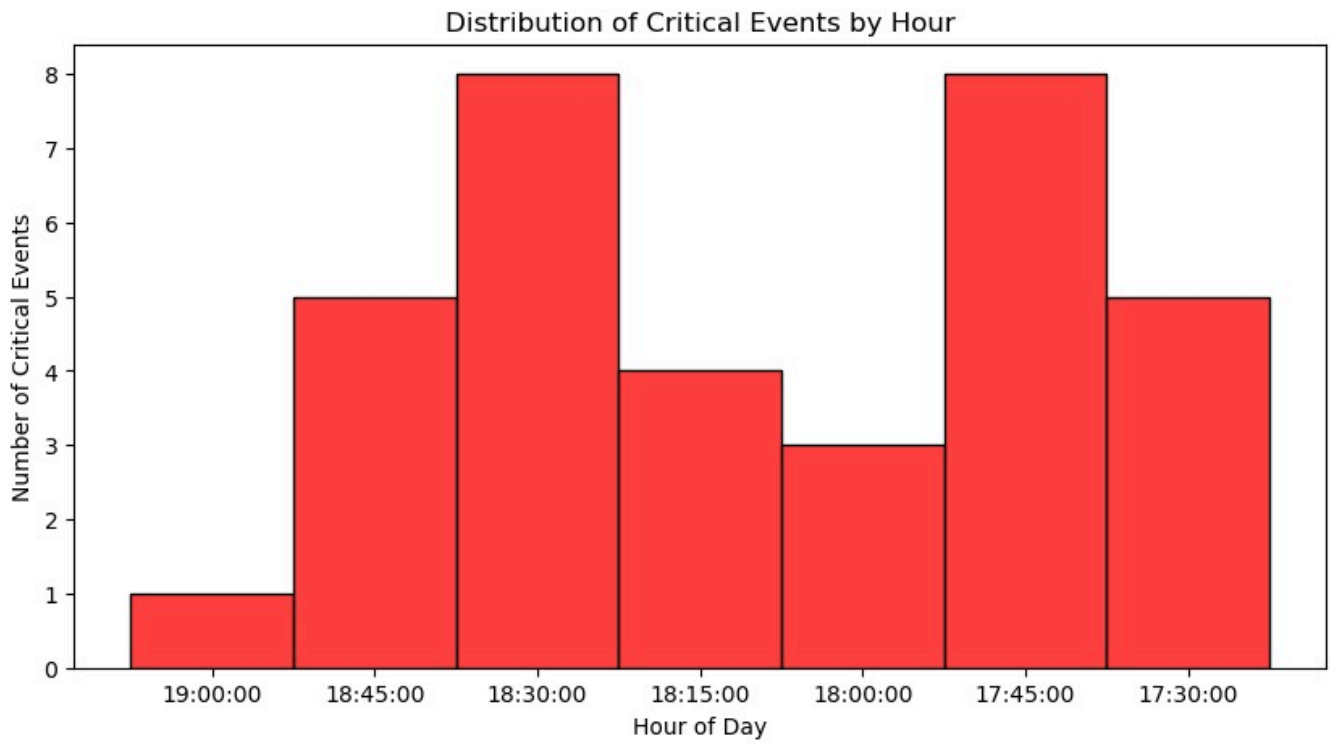


Figure 7.1.2: Distribution of detected grid stress events by hour of the day.

Key Insights

• **Evening Concentration of Stress Events:** Most critical events occur during the late afternoon and early evening hours (around 17:30–19:00).

• **Peak Grid Stress Window:** The clustering of events during this period indicates a consistent stress window, likely associated with increasing demand and declining renewable generation.

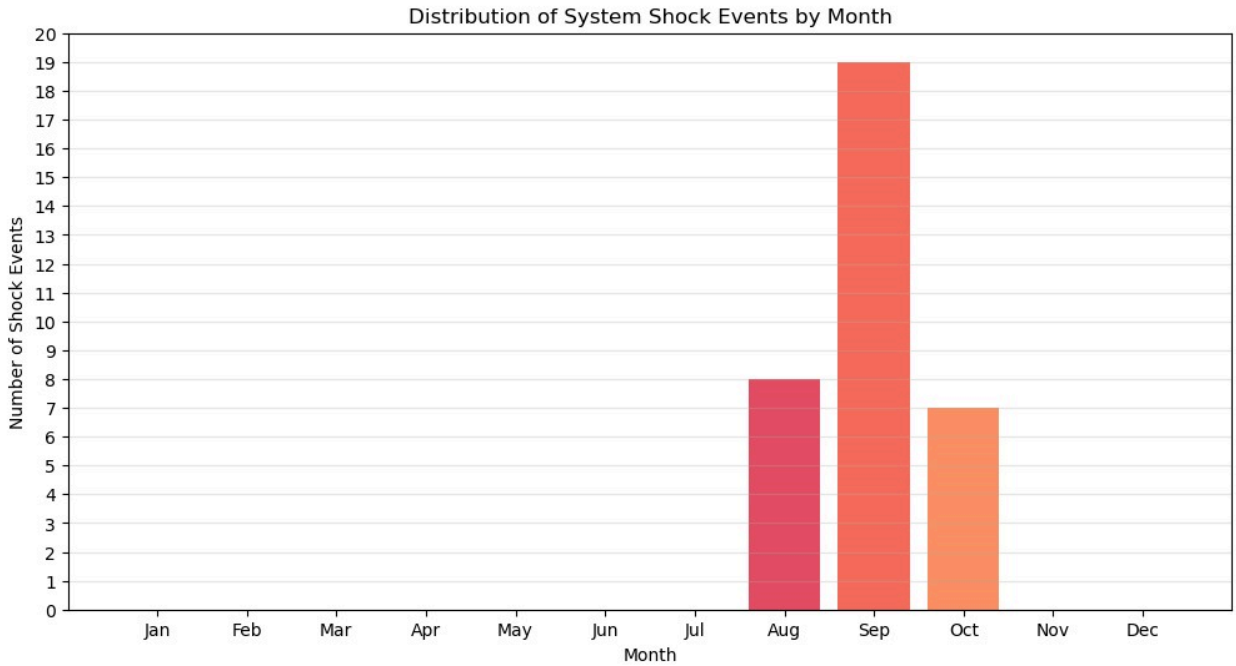


Figure 7.1.3: Monthly distribution of detected grid stress events in the California grid dataset.

Key Insights

• **Strong Summer Concentration:** Most grid stress events occur between August and October, indicating a strong seasonal pattern in system stress conditions.

• **Peak Stress Month:** September shows the highest number of detected events, suggesting this period is highly vulnerable to grid stress.

• **Extreme Climate Conditions:** California often experiences heatwaves and forest fires during these months, which can explain increasing electricity demand, and additional pressure on grid infrastructure.

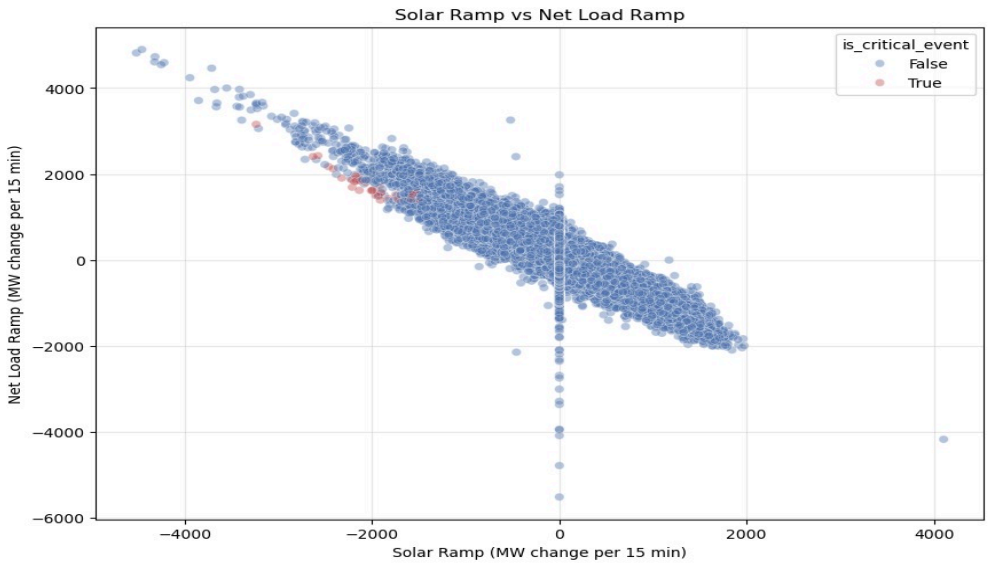


Figure 7.1.4: Relationship between solar generation ramp and net load ramp during grid operations.

Key Insights

• **Strong Negative Relationship:** The plot shows a clear inverse relationship between solar ramp and net load ramp, indicating that decreases in solar generation leads to sharp increases in net load.

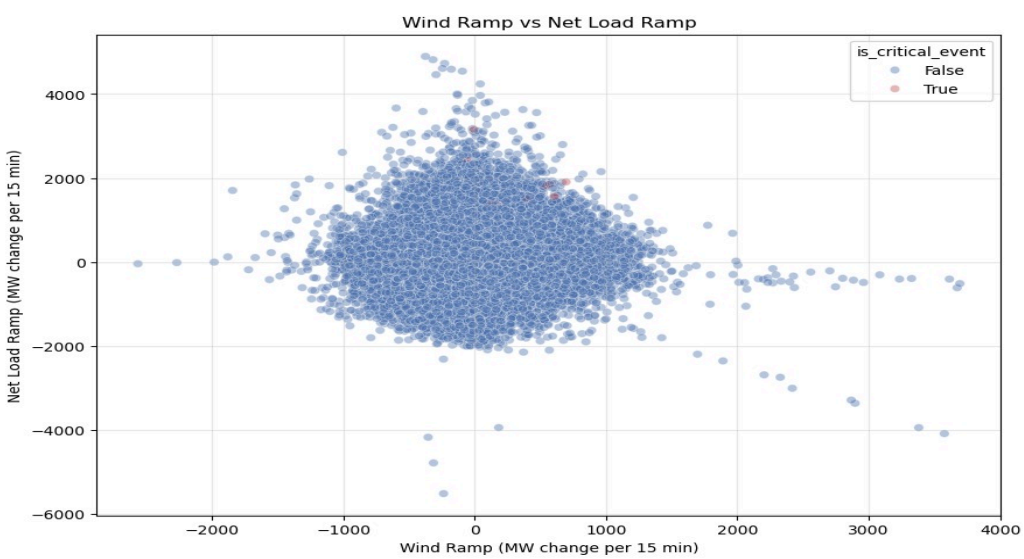


Figure 7.1.5: Relationship between wind generation ramp and net load ramp during grid operations.

Key Insights

• **Weak Correlation Compared to Solar:** Unlike solar generation, wind ramp changes show no strong linear relationship with net load ramp, indicating that wind variability alone does not consistently drive large net load changes.

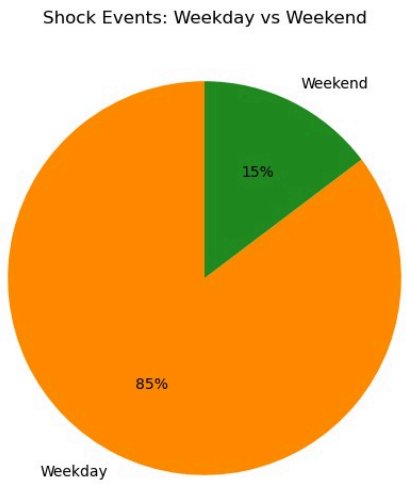


Figure 7.1.6: Distribution of detected grid stress events between weekdays and weekends.

Key Insights

• **Weekday Dominance:** Approximately 85% of the detected grid stress events occur on weekdays, indicating that system stress is significantly more frequent during working days due to industrial, commercial, and institutional activities, which can place additional pressure on the grid.

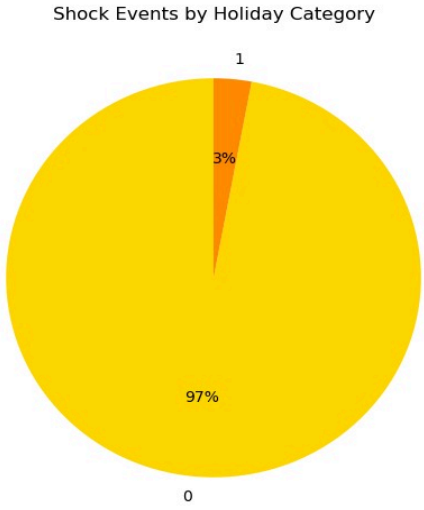


Figure 7.1.7: Distribution of detected grid stress events across holiday categories.

Key Insights

• **Predominance of Normal Working Days:** Approximately 97% of detected stress events occur on regular working days, indicating that grid stress is primarily associated with typical operational demand patterns.

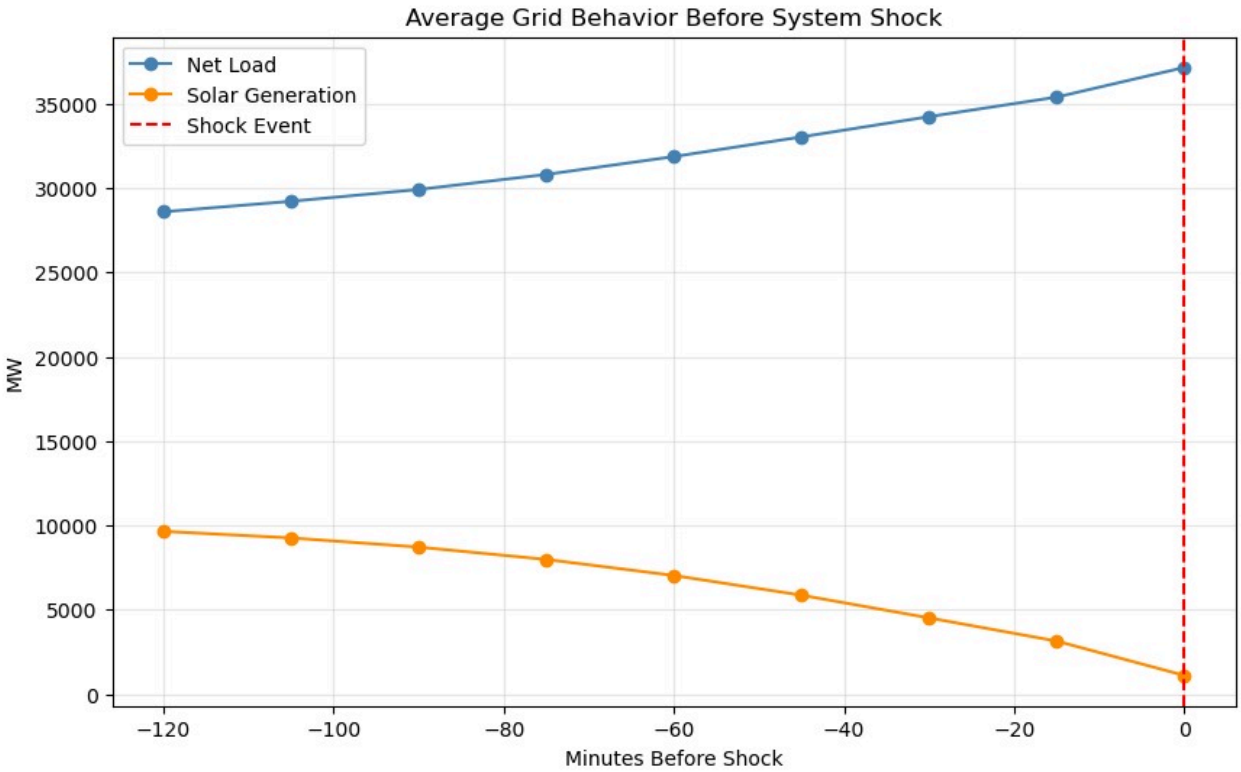


Figure 7.1.8: Average grid behavior in the two hours preceding a detected system shock event.

Key Insights

• **Gradual Net Load Increase:** Net load shows a steady rise in the two hours leading up to the shock event, indicating increasing demand or reduced renewable contribution as the system approaches stress conditions.

• **Rapid Solar Generation Decline:** Solar generation decreases sharply during the same period, suggesting that the reduction in solar output significantly contributes to rising net load.

• **Pre-Shock Stress Build-Up:** The observed trends indicate that grid stress events are typically preceded by a predictable build-up phase, where demand rises while renewable generation falls.

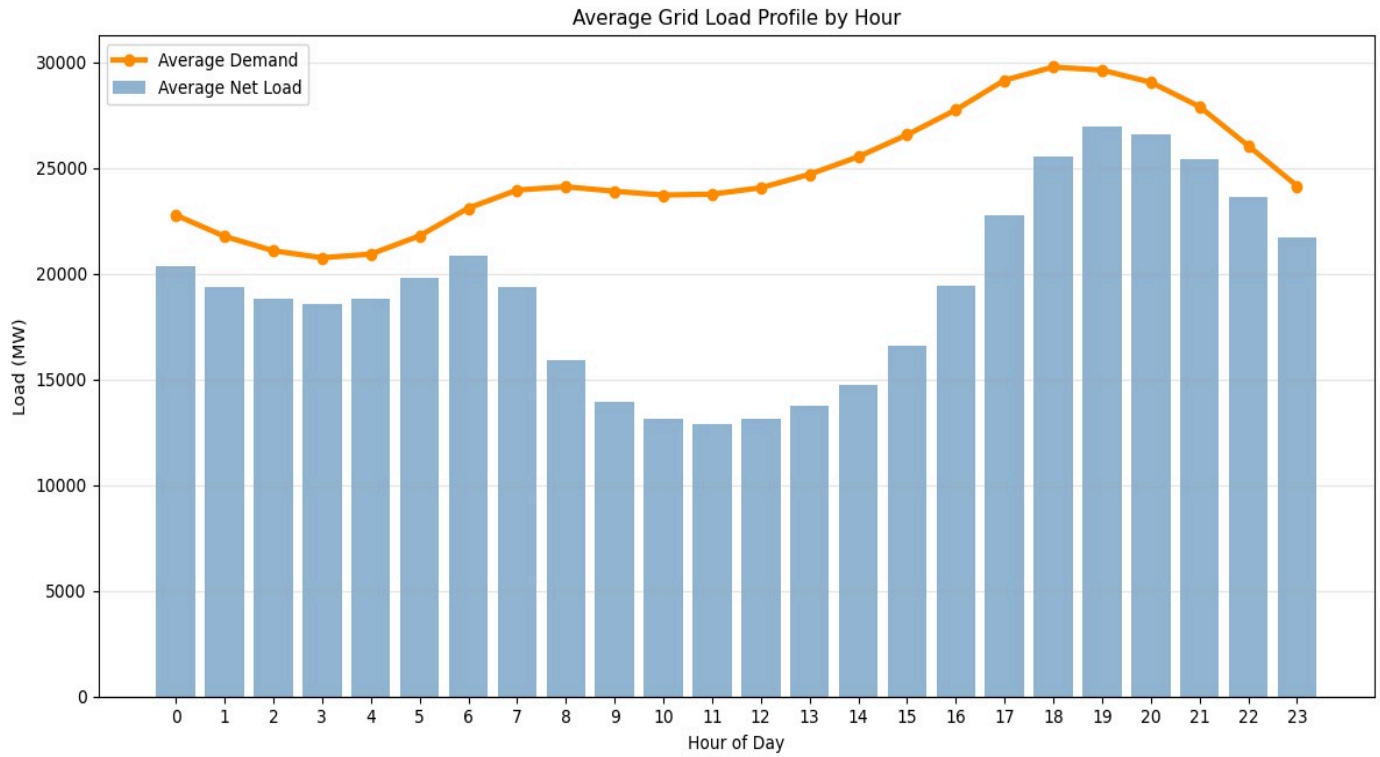


Figure 7.1.9: Average hourly grid demand and net load profile.

Key Insights

• **Evening Demand Surge:** Electricity demand gradually increases throughout the day and reaches its maximum during the evening hours (around 18:00–20:00).

• **Net Load Surge in Evening:** Net load also rises significantly during the evening period, indicating that the grid must supply more electricity from controllable generation sources.

• **Midday Renewable Offset:** During midday hours, net load remains comparatively lower despite rising demand, suggesting that solar generation helps offset a portion of the demand.

7.2 Grid Stress Detection using Machine Learning (Isolation Forest)

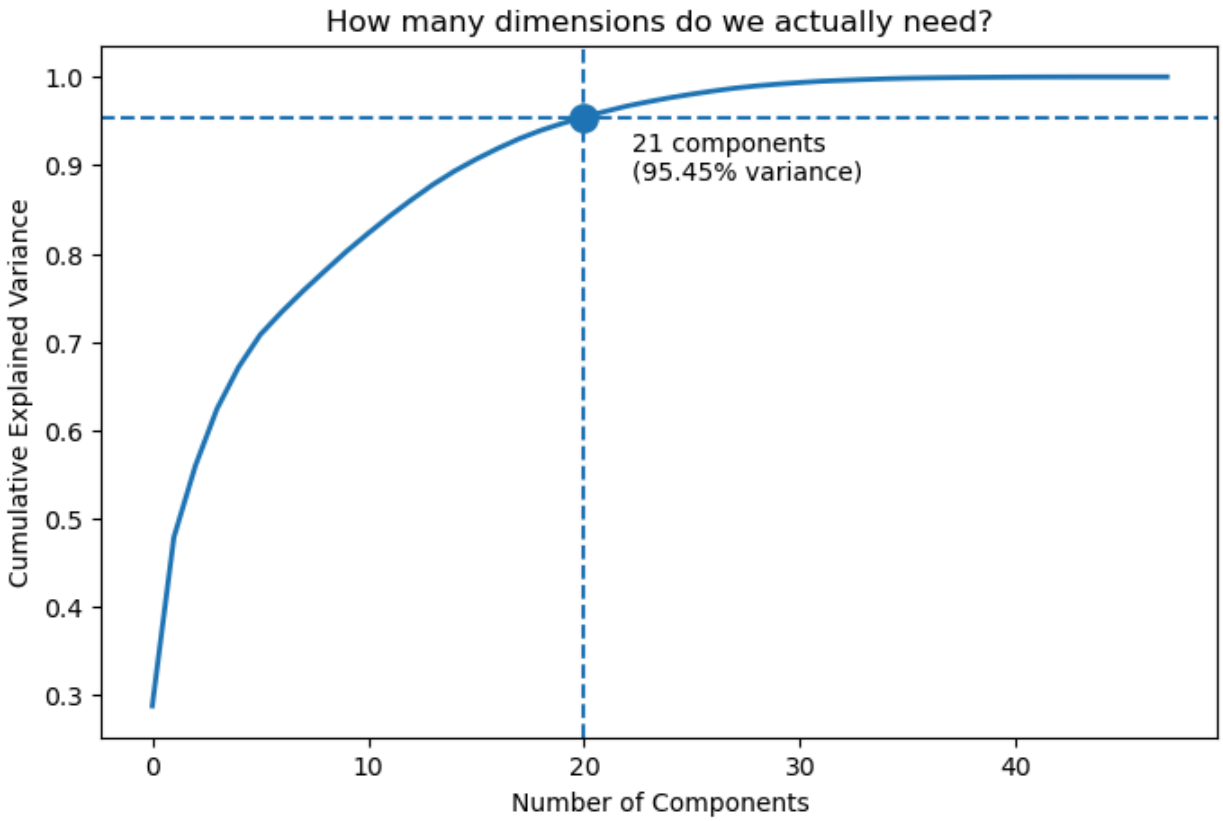


Figure 7.2.1: Cumulative explained variance from Principal Component Analysis (PCA) used for dimensionality reduction before applying Isolation Forest.

Key Insights

• **High dimensional redundancy:** The dataset originally contains 48 engineered features, many of which are correlated.

• **Dimensionality reduction efficiency:** PCA analysis shows that 21 components explain approximately 95.45% of the total variance, indicating that the majority of system information can be captured with fewer dimensions.

• **Model efficiency improvement:** Reducing the feature space helps improve Isolation Forest computational efficiency while preserving most of the system variability information.

Table 7.2.1: Comparison of average feature values during normal grid operation and Isolation Forest detected anomalies.

Metric	Normal Average	Anomaly Average	Percentage Change
Demand Ramp (mw)	-8.58	163.13	+2000%
Net Load Ramp (mw)	-24.94	473.76	+1999%
Solar Ramp (mw)	18.85	-358.06	-2000%
Wind Ramp (mw)	-9.92	188.91	+2004%
Heat Index	30.94	29.55	-4.5%
Demand Volatility (6h)	1514	2438	+64%

Key Insights

• **Ramp surge:** Both demand ramp and net load ramp increase by nearly 2000% during anomaly periods, indicating sudden and large system changes.

• **Solar generation collapse:** Solar ramp shifts from positive to strongly negative values, suggesting that rapid solar generation drops contribute to grid stress conditions.

• **Wind variability:** Wind ramp also shows large positive shifts during anomalies, indicating that renewable variability might increase system instability.

• **Demand volatility:** Demand volatility over a 6-hour window increases significantly, indicating higher short-term fluctuations in electricity demand during stress events.

• **Temperature impact:** Heat index shows only minor variation, suggesting that grid stress in these cases is not strongly associated with immediate temperature changes.

7.3 Frontend Interface and Visualization Snapshots

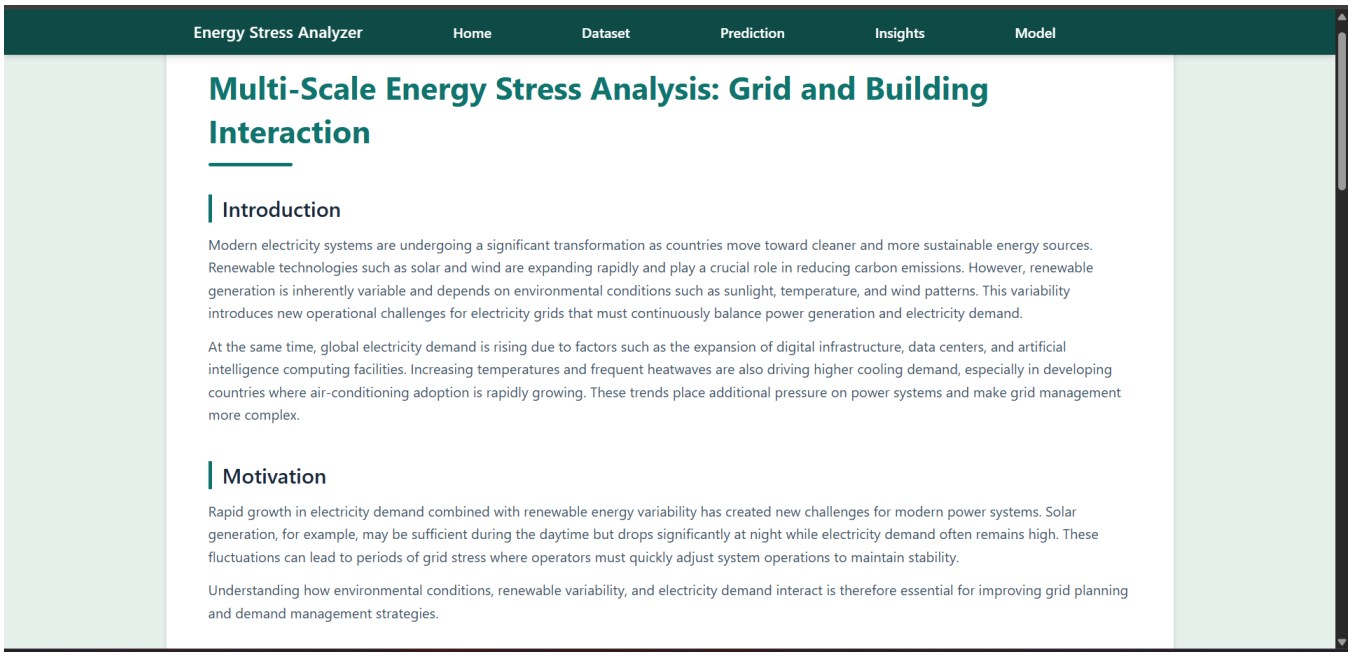
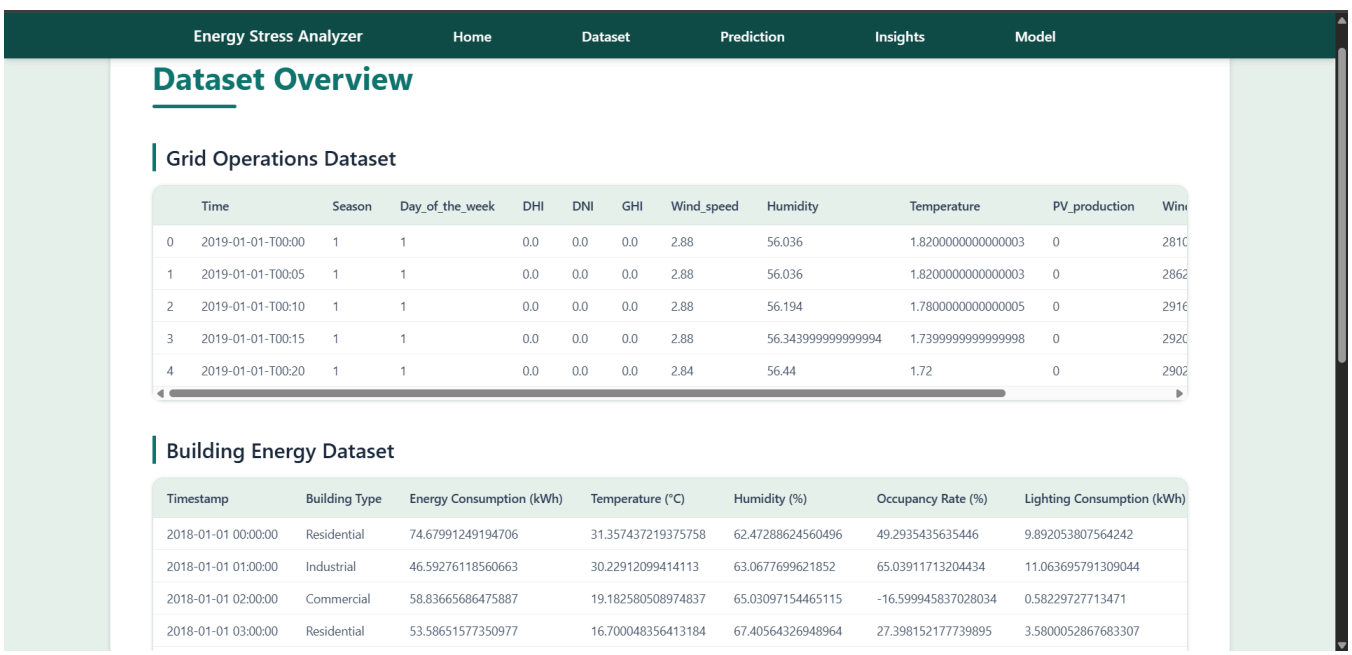


Figure 7.3.1: Home Page



	Time	Season	Day_of_the_week	DHI	DNI	GHI	Wind_speed	Humidity	Temperature	PV_production	Win
0	2019-01-01-T00:00	1	1	0.0	0.0	0.0	2.88	56.036	1.8200000000000003	0	2810
1	2019-01-01-T00:05	1	1	0.0	0.0	0.0	2.88	56.036	1.8200000000000003	0	2862
2	2019-01-01-T00:10	1	1	0.0	0.0	0.0	2.88	56.194	1.7800000000000005	0	2916
3	2019-01-01-T00:15	1	1	0.0	0.0	0.0	2.88	56.343999999999994	1.7399999999999998	0	2920
4	2019-01-01-T00:20	1	1	0.0	0.0	0.0	2.84	56.44	1.72	0	2902

Timestamp	Building Type	Energy Consumption (kWh)	Temperature (°C)	Humidity (%)	Occupancy Rate (%)	Lighting Consumption (kWh)
2018-01-01 00:00:00	Residential	74.67991249194706	31.357437219375758	62.47288624560496	49.2935435635446	9.892053807564242
2018-01-01 01:00:00	Industrial	46.59276118560663	30.22912099414113	63.0677699621852	65.03911713204434	11.063695791309044
2018-01-01 02:00:00	Commercial	58.83665686475887	19.182580508974837	65.03097154465115	-16.599945837028034	0.58229727713471
2018-01-01 03:00:00	Residential	53.58651577350977	16.700048356413184	67.40564326948964	27.398152177739895	3.5800052867683307

Figure 7.3.2: Dataset Overview Page

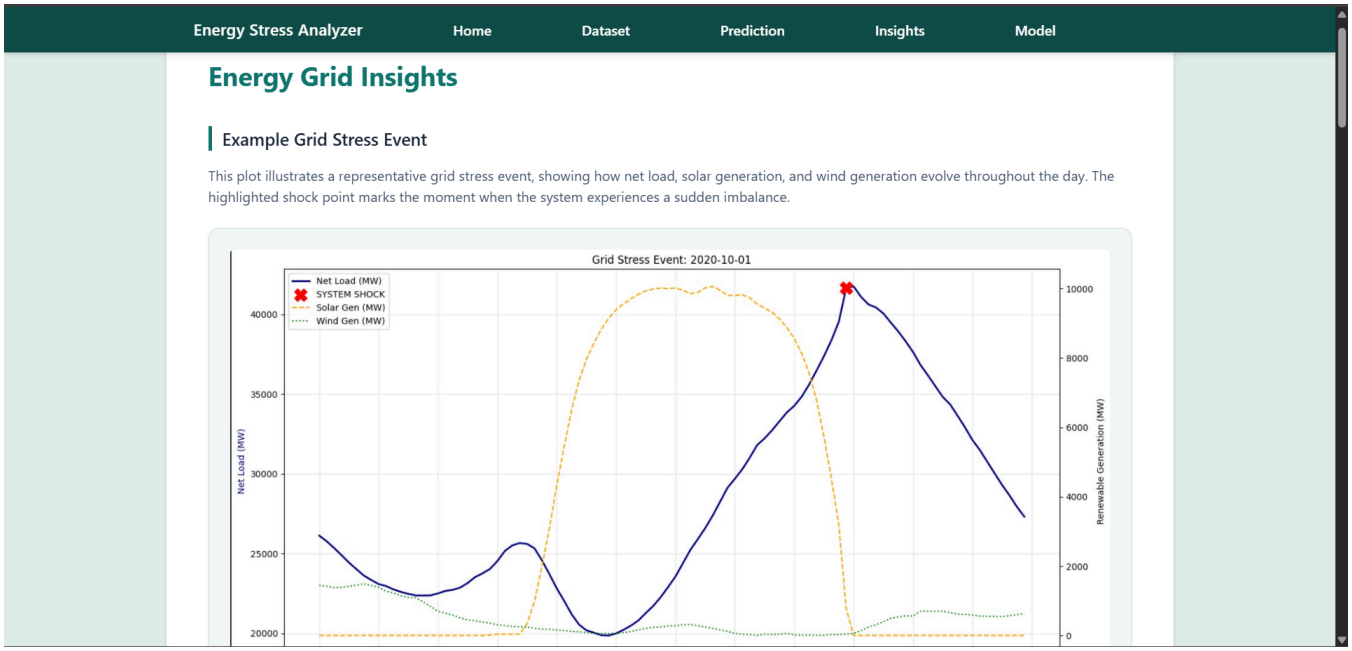


Figure 7.3.3: Insight Page

8. CONCLUSION

This project proposes a multi-scale analytical framework to study energy stress conditions in modern power systems by integrating grid operational data with building-level energy behavior. As renewable energy penetration increases, variability in generation and rising electricity demand create new challenges for maintaining grid stability.

By applying statistical analysis, anomaly detection methods, and machine learning models, the framework aims to identify abnormal grid behavior and analyze the factors that contribute to system stress. The project also explores how building-level energy consumption patterns interact with grid dynamics, providing a broader perspective on demand-side influences.

The integration of predictive modeling and anomaly detection enables the identification of potential high-risk periods and the development of early indicators of grid stress. Such insights can contribute to improved monitoring, planning, and management of electricity systems in regions experiencing rapid energy transition.

Overall, the proposed approach demonstrates how data-driven analysis can enhance the understanding of complex interactions between renewable generation, electricity demand, and building energy behavior within modern power grids.

9. REFERENCES

[1] A. Amllal, "California Power Consumption Dataset," Kaggle Dataset, 2023. [Online]. Available: https://www.kaggle.com/datasets/amineamllal/california-poc

[2] Dataset Engineer, "Southern California Energy Consumption Dataset," Kaggle Dataset, 2023. [Online]. Available: https://www.kaggle.com/datasets/datasetengineer/southern-california-energy-consumption

[3] A. Patel, "Feature Engineering and Feature Selection," GitHub Repository. Available: https://github.com/ashishpatel26/Amazing-Feature-Engineering

[4] H. Talsaniya, "Anomaly Detection with Unsupervised Machine Learning," Simform Engineering (Medium), Dec. 2023. [Online]. Available: https://medium.com/simform-engineering/anomaly-detection-with-unsupervised-machine-learning-3bcf4c431aff

[5] J. Nyangon and R. Akintunde, "Anomaly Detection in California Electricity Price Forecasting: Enhancing Accuracy and Reliability Using Principal Component Analysis," arXiv preprint arXiv:2412.07787, 2024.

[6] A. R. Kalair et al., "Duck Curve Leveling in Renewable Grids," Journal of Cleaner Production.